



Addressing challenges of real-world problems with **MULTIOBJECTIVE OPTIMIZATION**

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Motivation

Maximize the safety features

Maximize the engine power

Maximize the luxury



Which car
should I buy?

Minimize cost

Minimize fuel consumption

Minimize air pollution

Motivation

A supply chain manager
in a company ...

... needs to decide optimal quantity of item(s) to be ordered in each period,
in order to:



Minimize costs

Minimize storage
usage

Minimize
environment impact

Lot Sizing Problems

Maximize demand
satisfaction

Maximize inventory
turn over

Maximize profit

Minimize
financial risk

Multiobjective Optimization Problem (MOP)

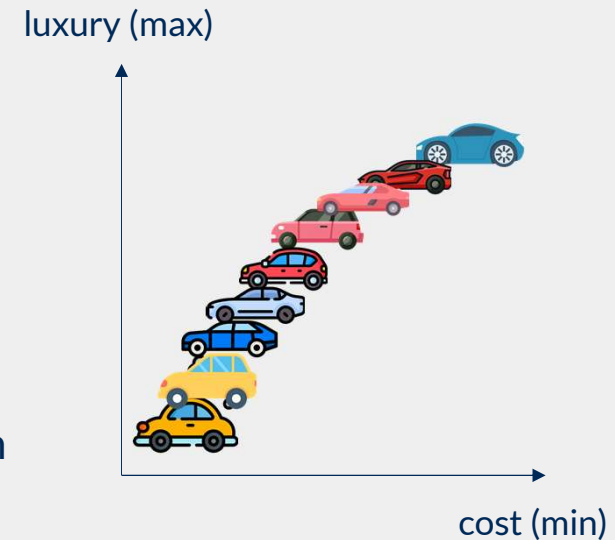
$$\begin{aligned} &\text{minimize } \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x}))^T \\ &\text{subject to } \mathbf{x} \in S \end{aligned}$$

- A **decision variable** vector $\mathbf{x} = (x_1, x_2, \dots, x_n)$ are **feasible** if they belong to the feasible region S which is a subset of the **decision space** $\mathbb{R}^n$. The feasible region is formed by constraints, which can be equality and inequality constraints and/or lower and upper bounds for the decision variable vectors.
- For a feasible decision variable vector $\mathbf{x} \in S$, the corresponding vector $\mathbf{z} = \mathbf{f}(\mathbf{x})$ is called a feasible **objective vector**, which consists of objective function values calculated at $\mathbf{x}$.
- The feasible objective vector belongs to the feasible objective region $Z = \mathbf{f}(S)$, which is a subset of the **objective space** $\mathbb{R}^k$. In what follows, we use the term '**solution**' to refer to an objective vector.

Optimal solutions of MOP

$$\begin{aligned} &\text{minimize } \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_k(\mathbf{x}))^T \\ &\text{subject to } \mathbf{x} \in S \end{aligned}$$

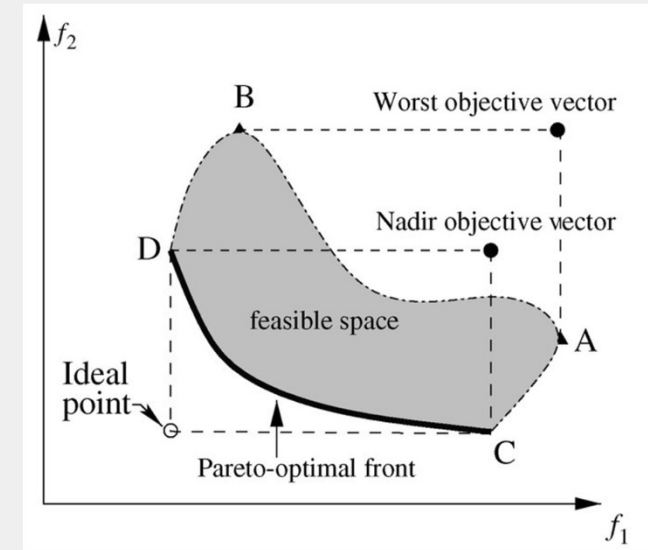
- An MOP has several objective functions to optimize simultaneously. Therefore, there is no single optimal solution where all objective functions achieve their individual optima.
- An MOP has several mathematically incomparable solutions, called Pareto optimal solutions
- A solution is Pareto optimal if none of the objectives can be improved without impairing at least one of the others.
- Solving an MOP can be understood as finding the most preferred solution for a decision maker (DM), who has expertise in the problem domain.



Optimal solutions of MOP

minimize $f(x) = (f_1(x), f_2(x), \dots, f_k(x))^T$
subject to $x \in S$

- Let z^1 and z^2 be two feasible solutions. We say that z^1 **dominates** z^2 if $z_i^1 \leq z_i^2$ for all $i = 1, \dots, k$ and $z_j^1 < z_j^2$ for at least one index j . The solutions that do not dominate each other are called **nondominated solutions**.
- A feasible solution $z' \in Z$ and its corresponding decision variable vector $x \in S$ are **Pareto optimal** if no other feasible solution $z \in Z$ dominates z' .
- The set of all Pareto optimal solutions is called a **Pareto optimal front**.
- To describe the ranges of objective function values in the Pareto optimal front, we define :
 - An **ideal point** : the best values that objective functions can achieve in the Pareto optimal front.
 - A **nadir point** : the worst values of objective functions in the Pareto optimal front.



Methods

To select one final solution to be applied in practice, we need preference information from a DM.

Based on the role of the DM in the decision making process, methods to solve MOPs are divided into four classes.

- **no-preference methods**, where no preference information from the DM is used;
- **a priori methods**, where the DM is first asked to specify his/her preferences, and a Pareto optimal solution that satisfies the preferences as well as possible is then found;
- **a posteriori methods**, where a (representative) set of Pareto optimal solutions is first generated and presented to the DM, and then he/she is expected to choose the most preferred one among them; and
- **interactive methods**, where the DM provides his/her preferences iteratively during the decision making process.

Interactive methods

Interactive
methods

In interactive methods,

- the decision maker is asked to **provide his/her preference information iteratively**.
- the decision maker is allowed to **learn different aspects of the problem** during the solution process.
- the decision maker is allowed to **change his/her preferences** until the most preferred solution is found.

Example of interactive methods : E-NAUTILUS¹, NAUTILUS Navigator², NIMBUS³

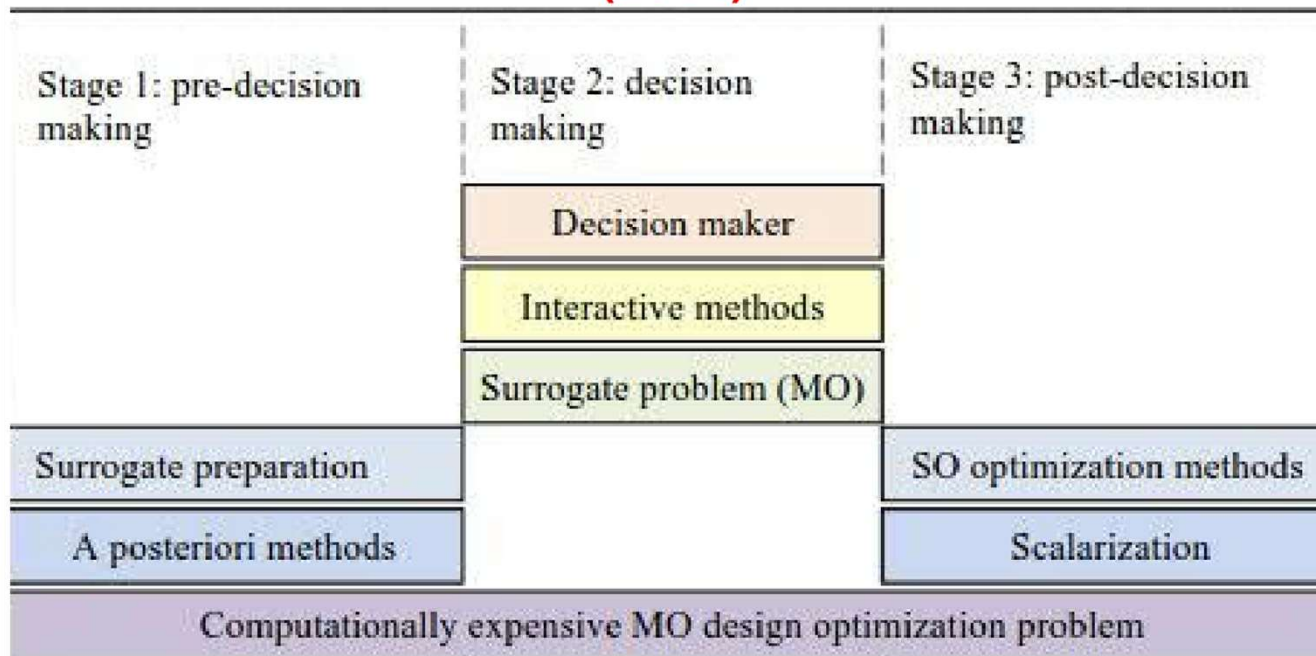


(1) Ruiz, A. B., et al. (2015). E-NAUTILUS: A decision support system for complex multiobjective optimization problems based on the NAUTILUS method. European Journal of Operational Research, 246(1), 218-231
(2) Ruiz, A. B., et al. (2019). NAUTILUS Navigator: free search interactive multiobjective optimization without trading-off. Journal of Global Optimization, 74(2), 213-231
(3) Miettinen, K., Mäkelä, M.M., Synchronous approach in interactive multiobjective optimization, European Journal of Operational Research, 170(3), 909-922, 2006

Computationally expensive problem

3-Stage Approach

Steponavice et al., Computer-Aided Design
(2014)





Traditional Methods

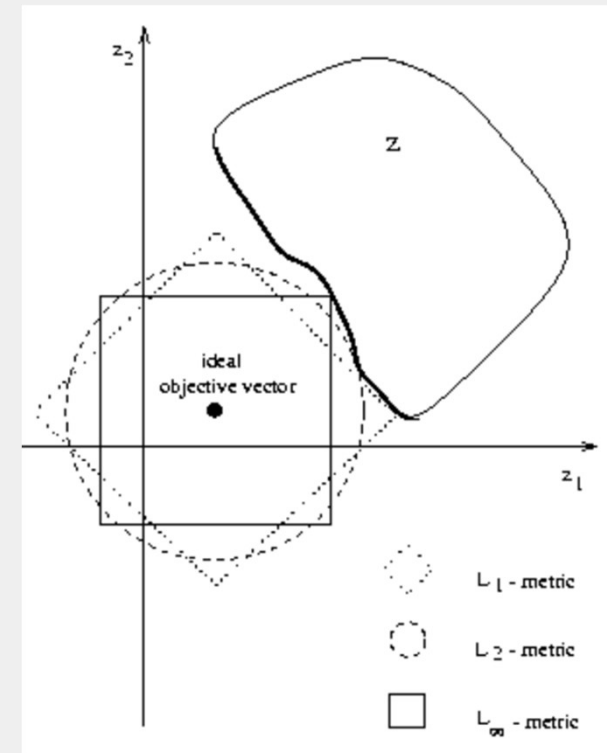


Global Criterion Method

The Global Criterion Method, also known as Compromise Programming, is a no-preference method. It does not require preference information from the DM. Instead, it seeks the Pareto-optimal solution closest to the ideal point by minimizing a distance measure.

$$\min_{x \in S} \left(\sum_{i=1}^k |f_i(x) - z_i^*|^p \right)^{1/p}, \quad x \in S.$$

The choice of p has a significant impact on the obtained solution. As shown in the figure, different distance metrics L_1 , L_2 , and L_∞ produce different contour shapes and may lead to different Pareto-optimal solutions.



Weighting method

The weighting method is one techniques to combine multiple objectives into a single-objective function by associating each objective function with a weighting coefficient and minimize the weighted sum of the objectives.

$$\text{minimize } \sum_{i=1}^k w_i f_i(\mathbf{x})$$

$$\text{subject to } \mathbf{x} \in S$$

$$\text{where } w_i \geq 0 \text{ for all } i = 1, \dots, k \text{ and } \sum_{i=1}^k w_i = 1$$

This method can be used as a priori methods by asking the DM to provide the weight of each objective as his/her preference, or as a posteriori methods by assigning several combination of weights and generate the solutions.

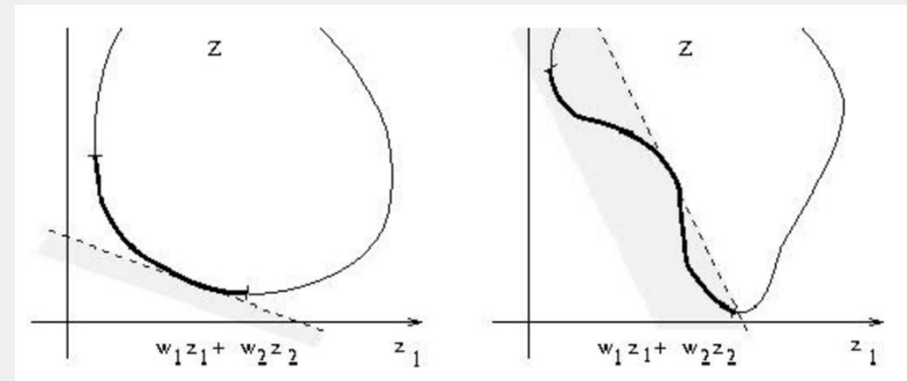
Weighting method



- Simplifies a complex multi-objective problem into a form that is solvable using standard optimization solvers.
- The Pareto optimality of the solution is guaranteed if $w_i > 0, i = 1, \dots, k$



- Some of the Pareto optimal solutions may fail to be found for nonconvex problem.
- Evenly distributed weights do not produce an evenly distributed representation of the Pareto optimal set.



ε -constraint method

The ε -constraint method is a technique where one objective function is chosen to be optimized, and the other objective functions are converted into constraints by setting upper bounds (ε -values) on them. This allows a multi-objective optimization problem to be handled using standard single-objective optimization techniques.

$$\begin{array}{ll}\text{minimize} & f_\ell(\mathbf{x}) \\ \text{subject to} & f_j(\mathbf{x}) \leq \varepsilon_j \quad \text{for all } j = 1, \dots, k, j \neq \ell \\ & \mathbf{x} \in S\end{array}$$

This method can be used as a priori methods by asking the DM to choose the one objective function to be optimized and define ε -values for the other objectives as his/her preference, or as a posteriori methods by assigning several combination of the ε -values and generate the solutions.

ϵ -constraint method



- Can generate Pareto-optimal solutions even for non-convex problems (with some efforts)
- The Pareto optimality of the solution is guaranteed if the solution is unique



- There may be difficulties in specifying which objective to be optimized and defining upper bounds for the other objectives
- The problem might become infeasible if constraints are too strict.

Problem in traditional a posteriori methods : We need to perform several runs with different parameter settings to achieve a set of Pareto optimal solutions.

Scalarizing Functions

A scalarizing function $s : \mathbb{R}^n \rightarrow \mathbb{R}$ transforms a multiobjective optimization problem into a single-objective optimization problem:

$$\begin{aligned} &\text{minimize } s(\mathbf{f}(\mathbf{x})) \\ &\text{subject to } \mathbf{x} \in S \end{aligned}$$

This transformation enables the use of single-objective optimization algorithms to solve multiobjective optimization problems. Scalarizing functions typically incorporate the DM's preferences, for example through a reference point.

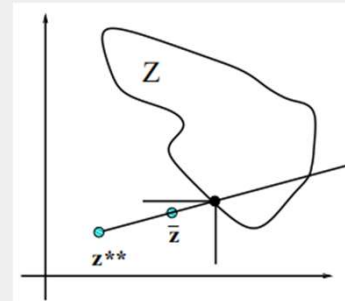
An ideal scalarizing function should satisfy:

- Every Pareto-optimal solution can be generated.
- Every obtained solution is Pareto optimal.
- If the aspiration levels are feasible, the obtained solution satisfies those aspiration levels.

Examples of Scalarizing Functions

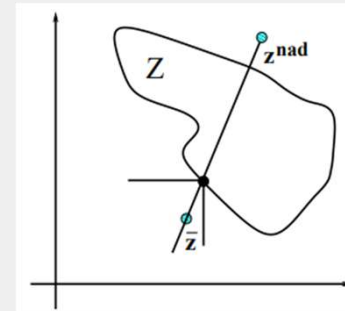
1. Satisficing Trade-Off Method (STOM)

$$s(\mathbf{f}(\mathbf{x})) = \max_i \left(\frac{f_i(\mathbf{x}) - z_i^{**}}{\bar{z}_i - z_i^{**}} \right)$$



2. GUESS scalarizing function

$$s(\mathbf{f}(\mathbf{x})) = \max_i \left(\frac{f_i(\mathbf{x}) - z_i^{nad}}{z_i^{nad} - \bar{z}_i} \right)$$



3. Achievement Scalarizing Function (ASF)

Finds the Pareto-optimal solution closest to the decision maker's aspiration levels.

$$s(\mathbf{f}(\mathbf{x})) = \max_i \left(\frac{f_i(\mathbf{x}) - \bar{z}_i}{z_i^{nad} - z_i^{**}} \right)$$



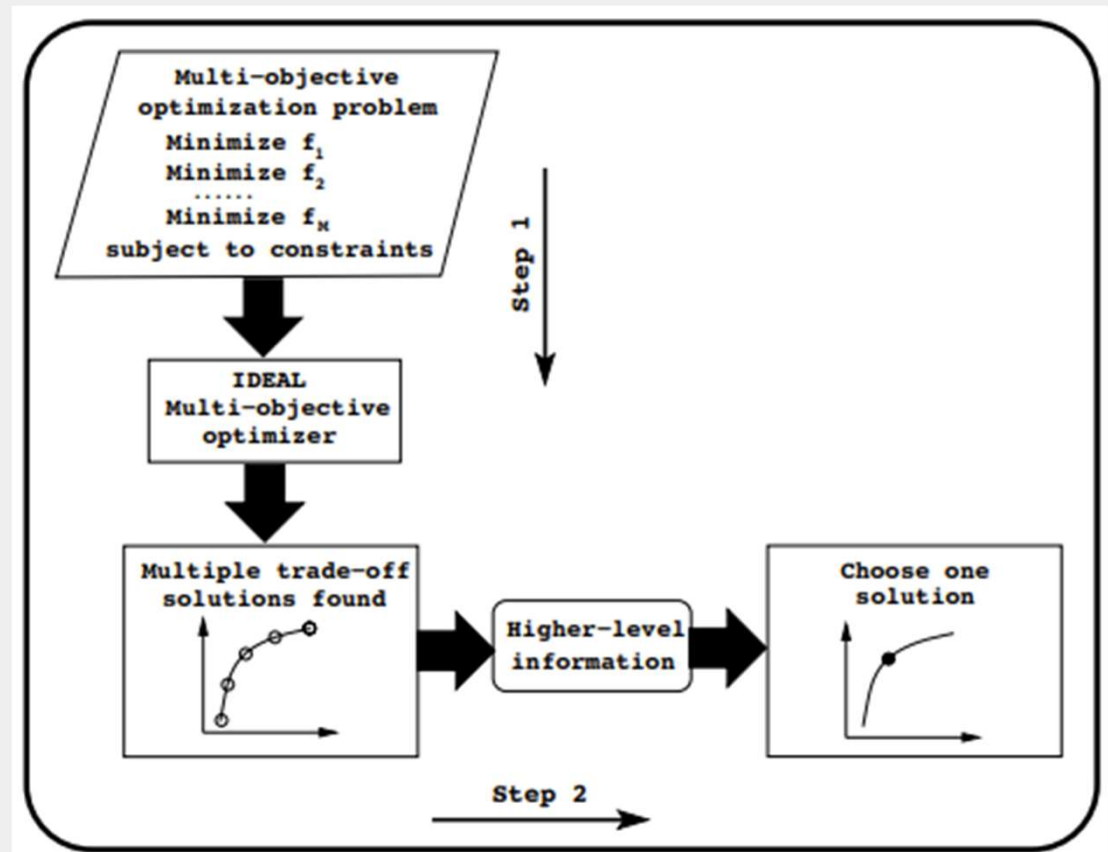
Evolutionary Multiobjective Optimization (EMO)



EMO as a posteriori methods

As a posteriori methods, EMO generates a representative set of (approximated) Pareto optimal solutions without DM preferences.

These solutions are then presented to the DM, and then he/she is expected to choose the most preferred one among them.



EMO goals

◆ 1. Proximity to the True Pareto Front (Convergence)

"The best-known Pareto front should be as close as possible to the true Pareto front."

◆ 2. Uniform Distribution (Diversity)

"Solutions... should be uniformly distributed and diverse... to provide the decision-maker a true picture of trade-offs."

◆ 3. Coverage of Entire Pareto Front (Extremeness)

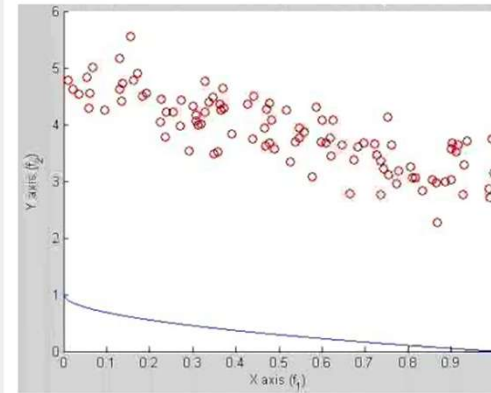
"Capture the whole spectrum... requires investigating solutions at the extreme ends."

EMO algorithm

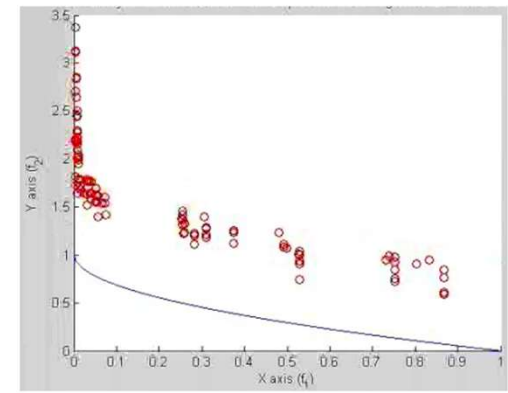
In EMO, we generate populations in the decision space. The corresponding solution (in the objective space) of the initial population may be far from the Pareto front.

As the algorithm progresses, solutions begin to converge toward the front.

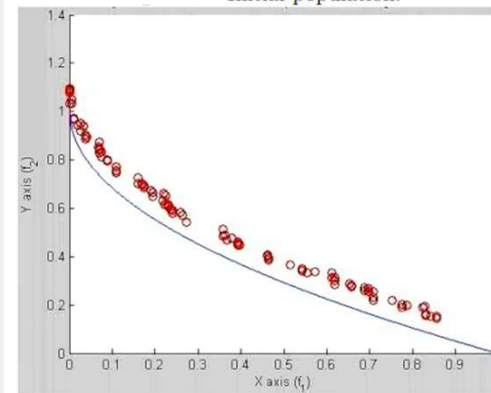
After several iterations, the population closely approximates the Pareto front with high convergence and good diversity, meaning the solutions are both close to the true front and evenly spread along it.



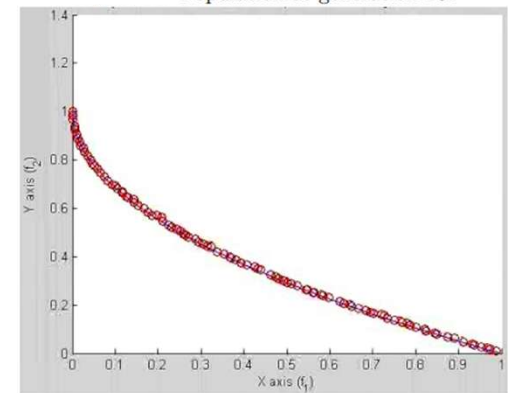
Initial population.



Population at generation 10.



Population at generation 30.



Population at generation 100.

Advantages and disadvantages of EMO

□ Advantages

- Works well for non-convex, non-differentiable, discontinuous, or noisy objective functions because EMO is derivative-free.
- Enable of finding several members of Pareto optimal set in a single run of the algorithm.
- Explore solutions over the entire search space

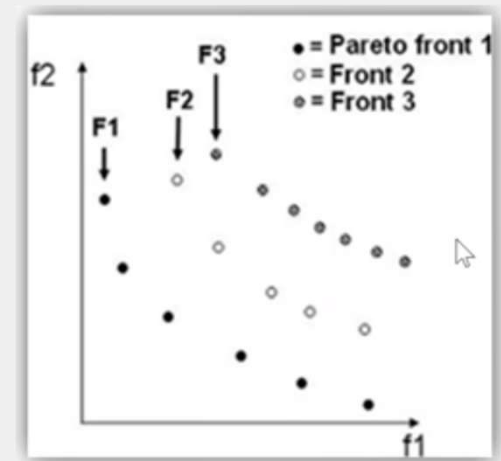
□ Disadvantages

- EMO is metaheuristic-based, so it does not guarantee finding the true Pareto front, only an approximation.

Nondominated Sorting Genetic Algorithm (NSGA) II

□ Non-dominated Sorting

- NSGA-II ranks the population into multiple fronts based on Pareto dominance.
- Front 1 consists of all non-dominated individuals.
- Front 2 consists of individuals dominated only by those in front 1, and so on.
- This allows the algorithm to distinguish how “good” a solution is with respect to others.

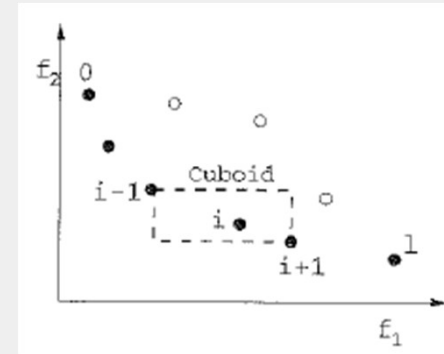


Nondominated Sorting Genetic Algorithm II

❑ Crowding Distance

$$CD_i = \sum_{m=1}^M \frac{f_m^{i+1} - f_m^{i-1}}{f_m^{\max} - f_m^{\min}}$$

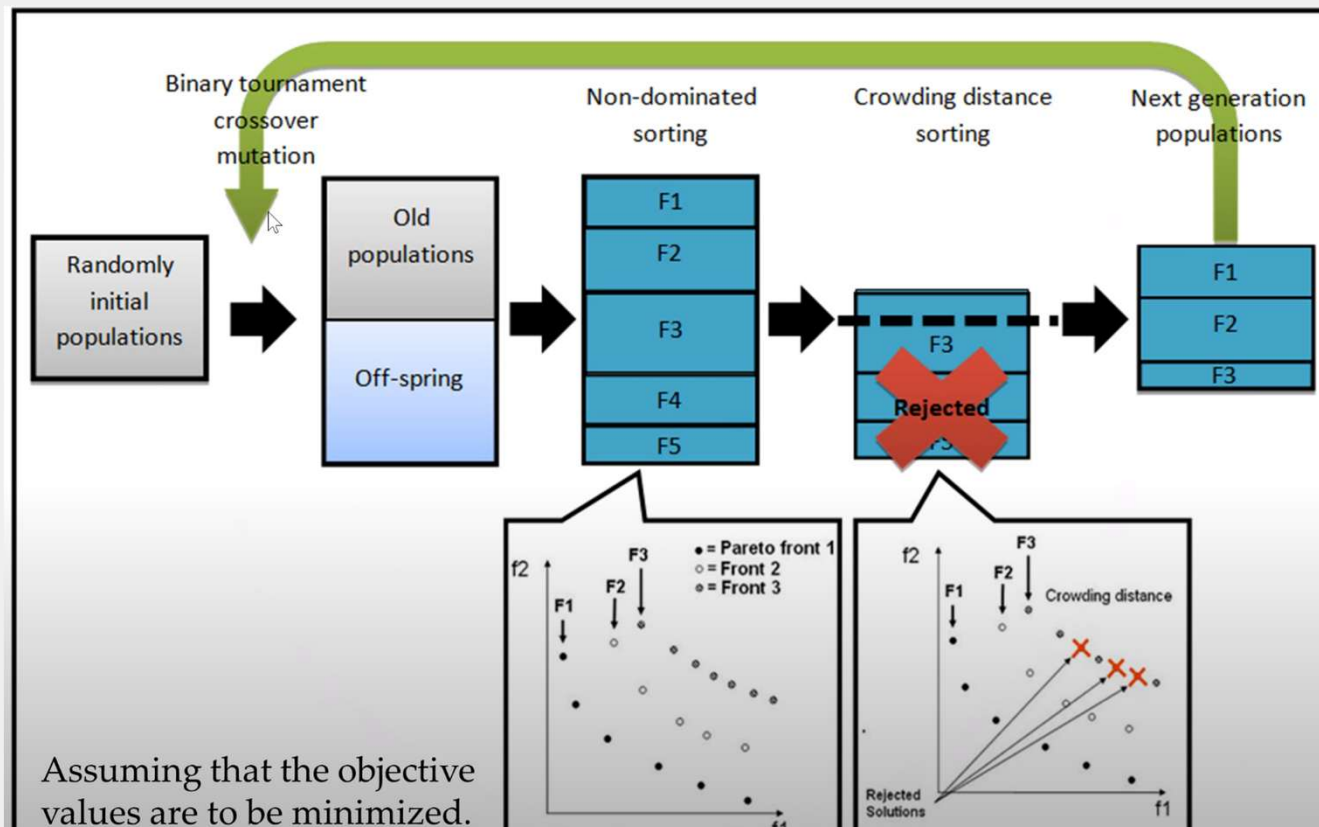
- Measures how close an individual is to its neighbors in the objective space.
- Encourages diversity by favoring individuals in less crowded regions.



❑ Elitism (Preservation of Good Solutions)

- Combines parents and offspring into a single pool of size 2N.
- Then, selects the best N solutions for the next generation based on: non-dominated rank and crowding distance
- This ensures that the best solutions are not lost over generations.

Nondominated Sorting Genetic Algorithm II





Interactive Methods



Interactive Methods

- In interactive methods, the DM is assumed to have knowledge about the problem in question, no deep understanding of optimization or its theory.
- Solution pattern is formed and repeated iteratively
- Role of DM is important. The DM is actively steering the solution process. The final solution is chosen by the DM, based on their own preferences.
- Only some Pareto optimal solutions (those that are interesting to the DM) are generated.
- DM is not overloaded with information.
- DM can learn : specify and correct preferences as the solution process continues.
- Psychological convergences!

Core Algorithm

1. Initialize the solution process, e.g., calculate the ideal point and the nadir point.
2. Generate an initial (Pareto optimal) solution as a current solution.
3. Ask the DM to provide preference information related to the current solution.
4. Generate new solution(s) based on the preference information
5. Ask the DM to choose the best solution of the previously generated solutions and denote it as the current solution
6. Check if the DM is satisfied with the current solution then the process stops. Otherwise, go back to Step 3 to repeat the loop.

Preference Information

DM provides his/her preferences in different ways (based on the methods).

Here are some examples :

1. Reference point: A reference point is a vector of aspiration levels, which consists of the desirable values for each objective function given by the DM.
2. Classification: The DM indicates which objectives should be improved, which ones can be sacrificed or are satisfactory.
3. Selection one among some solutions: A set of Pareto optimal solutions is shown to the DM. The DM then selects the one he/she prefer most. This selection reflects implicit preferences and helps the algorithm generate better options.

NIMBUS Method

- NIMBUS is a Pareto based method where DM move from one Pareto optimal solution to the other one in each iteration.
- In NIMBUS, the DM classifies each objective function of the current solution into 5 classes according to how he/she wants them to change in the next solution.

Class Type	Description	Symbol
1. Should be decreased	The DM wants the value of this objective to go down as much as possible.	$i \in I^{\Downarrow}$
2. Should be decreased to a target	The DM wants the objective to go down, but only until it reaches a specific target (aspiration level) $\bar{z}_i^h$.	$i \in I^{\Leftarrow}$
3. Satisfactory	The DM is satisfied with the current value of this objective; no change is needed.	$i \in I^=$
4. Allowed to increase (within a limit)	This objective can increase a bit, but not more than an upper bound ε_i^h .	$i \in I^{\triangleright}$
5. Free to change	The DM has no opinion; this objective can increase or decrease.	$i \in I^{\circ}$

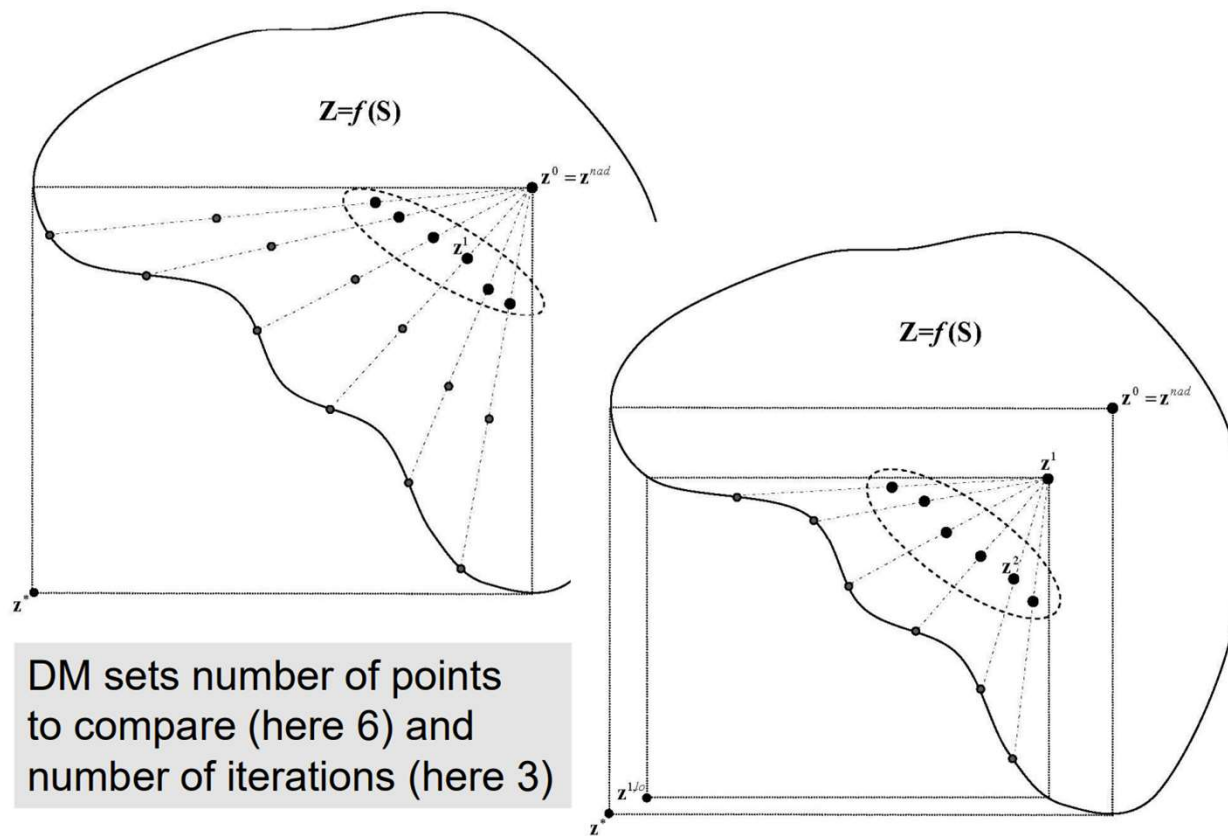
NIMBUS Method

1. Choose one Pareto Optimal solution as a starting point
2. Ask DM to classify objectives and define related parameters
3. Calculate 1-4 new Pareto Optimal solutions based on DM's preferences
4. Present these new solutions to the DM
5. If DM wants to save solutions, update database
6. Ask if DM wants intermediate solutions, then generate the intermediate solutions and present it to the DM. If DM wants to save the intermediate solutions, update database.
7. Ask DM to choose the most preferred solution

E – Nautilus Method

- Prospect Theory : “People do not respond equally to gains and losses; instead, they tend to fear losses more than desire gains”
- NAUTILUS methods are a trade-off free methods. These methods enable free search without requiring the DM to trade off.
- The algorithm typically starts at nadir point or another point where all objective values are poor. This allows clear room for improvement in every direction.
- At each iteration, the DM iteratively improves the solution — each new solution is better in all objectives than the previous one. In this way, the DM doesn't need to sacrifice one objective to improve another in the process.
- Only the final solution must be Pareto optimal

E – Nautilus Method



THANK YOU



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